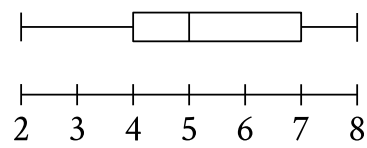


Question ID 57f45509

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 57f45509

PSAT1 M 2.1



The box plot summarizes **15** data values. What is the median of this data set?

- A. 2
- B. 3
- C. 5
- D. 8

ID: 57f45509 Answer

Correct Answer: C

Rationale

Choice C is correct. The median of a data set represented in a box plot is given by the vertical line within the box. In the given box plot, the vertical line within the box occurs at **5**. Therefore, the median of this data set is **5**.

Choice A is incorrect. This is the minimum value of the data set.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the maximum value of the data set.

Question Difficulty: Easy

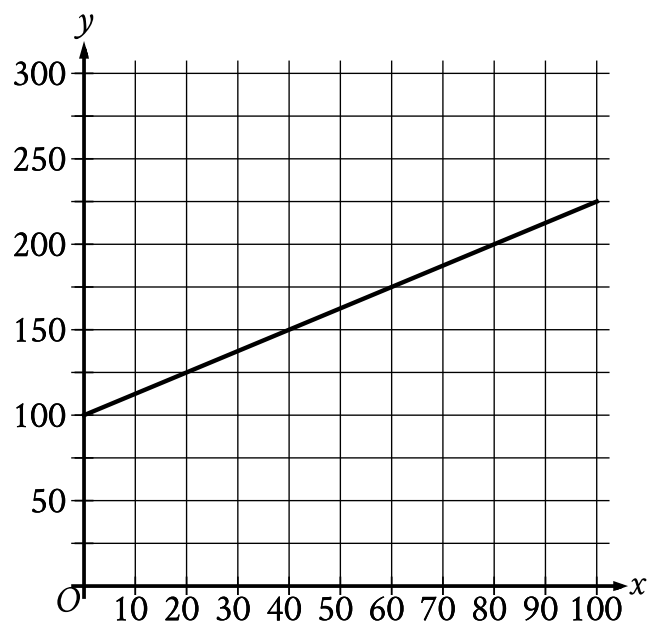
Question ID 720e51ac

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 720e51ac

PSAT1 M 2.2

The cost y , in dollars, for a manufacturer to make x rings is represented by the line shown.



What is the cost, in dollars, for the manufacturer to make **60** rings?

- A. 100
- B. 125
- C. 175
- D. 225

ID: 720e51ac Answer

Correct Answer: C

Rationale

Choice C is correct. The line shown represents the cost y , in dollars, for a manufacturer to make x rings. For the line shown, the x -axis represents the number of rings made by the manufacturer and the y -axis represents the cost, in dollars. Therefore, the cost, in dollars, for the manufacturer to make **60** rings is represented by the y -coordinate of the point on the line that has an x -coordinate of **60**. The point on the line with an x -coordinate of **60** has a y -coordinate of **175**. Therefore, the cost, in dollars, for the manufacturer to make **60** rings is **175**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the cost, in dollars, for the manufacturer to make **20** rings.

Choice D is incorrect. This is the cost, in dollars, for the manufacturer to make **100** rings.

Question Difficulty: Easy

Question ID b64e2c7f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: b64e2c7f

PSAT1 M 2.3

Monarch butterflies can fly only with a body temperature of at least **55.0 degrees Fahrenheit (°F)**. If a monarch butterfly's body temperature is **51.3°F**, what is the minimum increase needed in its body temperature, in **°F**, so that it can fly?

- A. **1.3**
- B. **3.7**
- C. **5.0**
- D. **6.3**

ID: b64e2c7f Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that monarch butterflies can fly only with a body temperature of at least **55.0 degrees Fahrenheit (°F)**. Let x represent the minimum increase needed in the monarch butterfly's body temperature to fly. If the monarch butterfly's body temperature is **51.3°F**, the inequality $51.3 + x \geq 55.0$ represents this situation. Subtracting **51.3** from both sides of this inequality yields $x \geq 3.7$. Therefore, if the monarch butterfly's body temperature is **51.3°F**, the minimum increase needed in its body temperature, in **°F**, so that it can fly is **3.7**.

Choice A is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is **53.7°F**, not **51.3°F**.

Choice C is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is **50.0°F**, not **51.3°F**.

Choice D is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is **48.7°F**, not **51.3°F**.

Question Difficulty: Easy

Question ID 837e9da7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 837e9da7

PSAT1 M 2.4

The function f is defined by $f(x) = \frac{1}{6x}$. What is the value of $f(x)$ when $x = 3$?

- A. $\frac{1}{3}$
- B. $\frac{1}{6}$
- C. $\frac{1}{9}$
- D. $\frac{1}{18}$

ID: 837e9da7 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that $f(x) = \frac{1}{6x}$. Substituting **3** for x in this equation yields $f(3) = \frac{1}{6(3)}$, or $f(3) = \frac{1}{18}$. Therefore, when $x = 3$, the value of $f(x)$ is $\frac{1}{18}$.

Choice A is incorrect. This is the value of $f(x)$ when $x = 0.5$.

Choice B is incorrect. This is the value of $f(x)$ when $x = 1$.

Choice C is incorrect. This is the value of $f(x)$ when $x = 1.5$.

Question Difficulty: Easy

Question ID b4553284

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: b4553284

PSAT1 M 2.5

If $2x = 12$, what is the value of $9x$?

ID: b4553284 Answer

Correct Answer: 54

Rationale

The correct answer is **54**. Dividing both sides of the given equation by **2** yields $x = 6$. Multiplying both sides of this equation by **9** yields $9x = 54$. Thus, the value of $9x$ is **54**.

Question Difficulty: Easy

Question ID c3989ef8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: c3989ef8

PSAT1 M 2.6

Henry receives a **\$60.00** gift card to pay for movies online. He uses his gift card to buy **3** movies for **\$7.50** each. If he spends the rest of his gift card balance on renting movies for **\$1.50** each, how many movies can Henry rent?

- A. 10
- B. 25
- C. 35
- D. 40

ID: c3989ef8 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that Henry uses his **\$60.00** gift card to buy **3** movies for **\$7.50** each. Therefore, Henry spends **3(\$7.50)**, or **\$22.50**, of his **\$60.00** gift card to buy **3** movies. After buying **3** movies with his **\$60.00** gift card, Henry has a gift card balance of **\$60.00 – \$22.50**, or **\$37.50**. It's also given that Henry spends the rest of his gift card balance on renting movies for **\$1.50** each. Therefore, Henry can rent $\frac{\$37.50}{\$1.50}$, or **25**, movies.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID ff2c1431

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: ff2c1431

PSAT1 M 2.7

$7m = 5(n + p)$

The given equation relates the positive numbers m , n , and p . Which equation correctly gives n in terms of m and p ?

- A. $n = \frac{5p}{7m}$
- B. $n = \frac{7m}{5} - p$
- C. $n = 5(7m) + p$
- D. $n = 7m - 5 - p$

ID: ff2c1431 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the equation $7m = 5(n + p)$ relates the positive numbers m , n , and p . Dividing both sides of the given equation by 5 yields $\frac{7m}{5} = n + p$. Subtracting p from both sides of this equation yields $\frac{7m}{5} - p = n$, or $n = \frac{7m}{5} - p$. It follows that the equation $n = \frac{7m}{5} - p$ correctly gives n in terms of m and p .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID bf883fde

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: bf883fde

PSAT1 M 2.8

For the function f , the graph of $y = f(x)$ in the xy -plane has a slope of 3 and passes through the point $(0, -8)$. Which equation defines f ?

- A. $f(x) = 3x$
- B. $f(x) = 3x - 8$
- C. $f(x) = 3x + 5$
- D. $f(x) = 3x + 11$

ID: bf883fde Answer

Correct Answer: B

Rationale

Choice B is correct. An equation defining a linear function can be written in the form $f(x) = mx + b$, where m and b are constants, m is the slope of the graph of $y = f(x)$ in the xy -plane, and $(0, b)$ is the y -intercept of the graph. It's given that for the function f , the graph of $y = f(x)$ in the xy -plane has a slope of 3 . Therefore, $m = 3$. It's also given that this graph passes through the point $(0, -8)$. Therefore, the y -intercept of the graph is $(0, -8)$, and it follows that $b = -8$. Substituting 3 for m and -8 for b in the equation $f(x) = mx + b$ yields $f(x) = 3x - 8$. Thus, the equation that defines f is $f(x) = 3x - 8$.

Choice A is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 0)$, not $(0, -8)$.

Choice C is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 5)$, not $(0, -8)$.

Choice D is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 11)$, not $(0, -8)$.

Question Difficulty: Easy

Question ID 79201024

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 79201024

PSAT1 M 2.9

A band with **45** members has **11** members who play saxophone. If one band member is selected at random, what is the probability of selecting a band member who plays saxophone?

- A. $\frac{1}{45}$
- B. $\frac{11}{45}$
- C. $\frac{34}{45}$
- D. $\frac{45}{45}$

ID: 79201024 Answer

Correct Answer: B

Rationale

Choice B is correct. The probability of an event occurring is the ratio of the number of favorable outcomes to the total number of possible outcomes. It's given that there are **45** band members, which is the total number of possible outcomes. It's also given that there are **11** band members who play saxophone. Therefore, the number of favorable outcomes is **11**. Thus, the probability of selecting a band member who plays saxophone is $\frac{11}{45}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the probability of selecting a band member who does not play saxophone.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 1e003284

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 1e003284

PSAT1 M 2.10

$$\begin{aligned}x &= 49 \\ y &= \sqrt{x} + 9\end{aligned}$$

The graphs of the given equations intersect at the point (x, y) in the xy -plane. What is the value of y ?

- A. 16
- B. 40
- C. 81
- D. 130

ID: 1e003284 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the graphs of the given equations intersect at the point (x, y) in the xy -plane. It follows that (x, y) represents a solution to the system consisting of the given equations. The first equation given is $x = 49$. Substituting 49 for x in the second equation given, $y = \sqrt{x} + 9$, yields $y = \sqrt{49} + 9$, which is equivalent to $y = 7 + 9$, or $y = 16$. It follows that the graphs of the given equations intersect at the point $(49, 16)$. Therefore, the value of y is 16.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

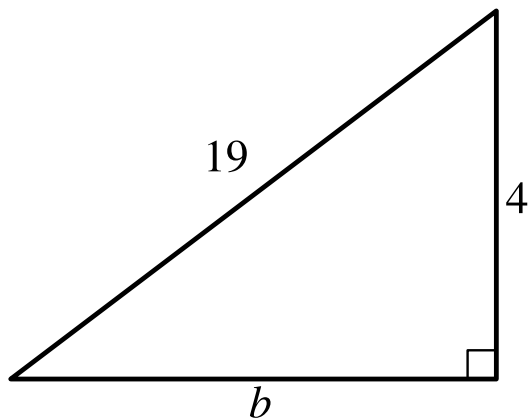
Question Difficulty: Easy

Question ID b0c5ece5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	■ □ □

ID: b0c5ece5

PSAT1 M 2.11



Note: Figure not drawn to scale.

Which equation shows the relationship between the side lengths of the given triangle?

- A. $4b = 19$
- B. $4 + b = 19$
- C. $4^2 + b^2 = 19^2$
- D. $4^2 - b^2 = 19^2$

ID: b0c5ece5 Answer

Correct Answer: C

Rationale

Choice C is correct. The Pythagorean theorem states that in a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. Therefore, $a^2 + b^2 = c^2$, where a and b are the lengths of the legs and c is the length of the hypotenuse. For the given right triangle, the lengths of the legs are 4 and b , and the length of the hypotenuse is 19 . Substituting 4 for a and 19 for c in the equation $a^2 + b^2 = c^2$ yields $4^2 + b^2 = 19^2$. Thus, the relationship between the side lengths of the given triangle is $4^2 + b^2 = 19^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID c8e0f511

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: c8e0f511

PSAT1 M 2.12

For a camping trip a group bought x one-liter bottles of water and y three-liter bottles of water, for a total of **240** liters of water. Which equation represents this situation?

- A. $x + 3y = 240$
- B. $x + y = 240$
- C. $3x + 3y = 240$
- D. $3x + y = 240$

ID: c8e0f511 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that for a camping trip a group bought x one-liter bottles of water and y three-liter bottles of water. Since the group bought x one-liter bottles of water, the total number of liters bought from x one-liter bottles of water is represented as $1x$, or x . Since the group bought y three-liter bottles of water, the total number of liters bought from y three-liter bottles of water is represented as $3y$. It's given that the group bought a total of **240** liters; thus, the equation $x + 3y = 240$ represents this situation.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This equation represents a situation where the group bought x three-liter bottles of water and y one-liter bottles of water, for a total of **240** liters of water.

Question Difficulty: Easy

Question ID 2d16d62c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 2d16d62c

PSAT1 M 2.13

A special camera is used for underwater ocean research. When the camera is at a depth of **58** fathoms, what is the camera's depth in feet? (**1 fathom = 6 feet**)

ID: 2d16d62c Answer

Correct Answer: 348

Rationale

The correct answer is **348**. It's given that **1** fathom is equivalent to **6** feet. Therefore, **58** fathoms is equivalent to **(58 fathoms)** $\left(\frac{6 \text{ feet}}{1 \text{ fathom}}\right)$, or **348** feet. Thus, when the camera is at a depth of **58** fathoms, the camera's depth, in feet, is **348**.

Question Difficulty: Easy

Question ID e7343559

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: e7343559

PSAT1 M 2.14

$$y = -4x + 40$$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
0	0
1	-4
2	-8

B.

x	y
0	40
1	44
2	48

C.

x	y
0	40
1	36
2	32

D.

x	y
0	0
1	4
2	8

ID: e7343559 Answer

Correct Answer: C

Rationale

Choice C is correct. Each of the given choices gives three values of x : 0, 1, and 2. Substituting 0 for x in the given equation yields $y = -4(0) + 40$, or $y = 40$. Therefore, when $x = 0$, the corresponding value of y for the given equation is 40. Substituting 1 for x in the given equation yields $y = -4(1) + 40$, or $y = 36$. Therefore, when $x = 1$, the corresponding value of y for the given equation is 36. Substituting 2 for x in the given

equation yields $y = -4(2) + 40$, or $y = 32$. Therefore, when $x = 2$, the corresponding value of y for the given equation is **32**. Choice C gives three values of x , **0**, **1**, and **2**, and their corresponding values of y , **40**, **36**, and **32**, respectively, for the given equation.

Choice A is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -4x$.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = 4x + 40$.

Choice D is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = 4x$.

Question Difficulty: Easy

Question ID f9d40000

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: f9d40000

PSAT1 M 2.15

In $\triangle XYZ$, the measure of $\angle X$ is 23° and the measure of $\angle Y$ is 66° . What is the measure of $\angle Z$?

- A. 43°
- B. 89°
- C. 91°
- D. 179°

ID: f9d40000 Answer

Correct Answer: C

Rationale

Choice C is correct. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is 180° . It's given that in $\triangle XYZ$, the measure of $\angle X$ is 23° and the measure of $\angle Y$ is 66° . It follows that the measure of $\angle Z$ is $(180 - 23 - 66)^\circ$, or 91° .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the sum of the measures of $\angle X$ and $\angle Y$, not the measure of $\angle Z$.

Choice D is incorrect and may result from conceptual or calculation errors.

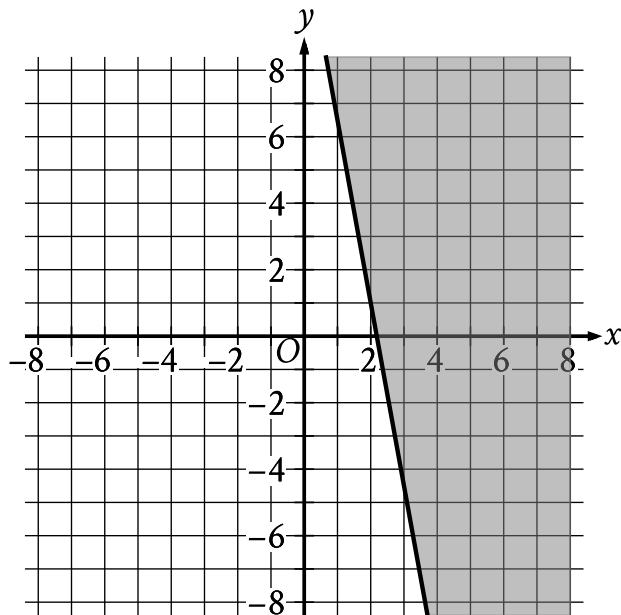
Question Difficulty: Easy

Question ID 59a49431

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	☒ ☐ ☐

ID: 59a49431

PSAT1 M 2.16



The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

- A. $(0, -4)$
- B. $(0, 4)$
- C. $(-4, 0)$
- D. $(4, 0)$

ID: 59a49431 Answer

Correct Answer: D

Rationale

Choice D is correct. Since the shaded region shown represents solutions to an inequality, an ordered pair (x, y) is a solution to the inequality if it's represented by a point in the shaded region. Of the given choices, only $(4, 0)$ is represented by a point in the shaded region. Therefore, $(4, 0)$ is a solution to the inequality.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID f88970cc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: f88970cc

PSAT1 M 2.17

$$\begin{aligned}x &= 5 \\ y &= x - 8\end{aligned}$$

Which of the following points (x, y) is the solution to the given system of equations in the xy -plane?

- A. $(0, 0)$
- B. $(5, -3)$
- C. $(5, -8)$
- D. $(5, 8)$

ID: f88970cc Answer

Correct Answer: B

Rationale

Choice B is correct. A solution to a system of equations in the xy -plane is a point (x, y) that lies on the graph of each equation in the system. The first equation given is $x = 5$. Substituting 5 for x in the second given equation yields $y = 5 - 8$, or $y = -3$. It follows that in the xy -plane, the point $(5, -3)$ lies on the graph of each equation in the system. Therefore, the solution to the given system of equations in the xy -plane is $(5, -3)$.

Choice A is incorrect. The point $(0, 0)$ doesn't lie on the graph of either equation in the given system.

Choice C is incorrect. The point $(5, -8)$ doesn't lie on the graph of the second equation in the given system.

Choice D is incorrect. The point $(5, 8)$ doesn't lie on the graph of the second equation in the given system.

Question Difficulty: Easy

Question ID c7bed21d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c7bed21d

PSAT1 M 2.18

Quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P , Q , R , and S correspond to P' , Q' , R' , and S' , respectively. The measure of angle P is 30° , the measure of angle Q is 50° , and the measure of angle R is 70° . The length of each side of $P'Q'R'S'$ is 3 times the length of each corresponding side of $PQRS$. What is the measure of angle P' ?

- A. 10°
- B. 30°
- C. 40°
- D. 90°

ID: c7bed21d Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P , Q , R , and S correspond to P' , Q' , R' , and S' , respectively. Since corresponding angles of similar quadrilaterals are congruent, it follows that the measure of angle P is equal to the measure of angle P' . It's given that the measure of angle P is 30° . Therefore, the measure of angle P' is 30° .

Choice A is incorrect. This is $\frac{1}{3}$ the measure of angle P' .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is 3 times the measure of angle P' .

Question Difficulty: Medium

Question ID 5ae186b4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 5ae186b4

PSAT1 M 2.19

$$\frac{-54}{w} = 6$$

What is the solution to the given equation?

ID: 5ae186b4 Answer

Correct Answer: -9

Rationale

The correct answer is -9 . Since w is in the denominator of a fraction in the given equation, w can't be equal to 0 . Since w isn't equal to 0 , multiplying both sides of the given equation by w yields an equivalent equation, $-54 = 6w$. Dividing both sides of this equation by 6 yields $-9 = w$. Therefore, -9 is the solution to the given equation.

Question Difficulty: Medium

Question ID d693f563

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: d693f563

PSAT1 M 2.20

Last year, Cedric had **35** plants in his garden. This year, the number of plants in Cedric’s garden is **60%** greater than the number of plants in his garden last year. How many plants does Cedric have in his garden this year?

ID: d693f563 Answer

Correct Answer: 56

Rationale

The correct answer is **56**. It’s given that Cedric had **35** plants in his garden last year and that the number of plants in Cedric’s garden this year is **60%** greater than the number of plants in his garden last year. It follows that the number of plants in Cedric’s garden this year is **35** plus **60%** of **35**, which is equal to **$35 + 35\left(\frac{60}{100}\right)$** , or **$35 + 35(0.6)$** . This expression is equivalent to **$35 + 21$** , or **56**. Therefore, Cedric has **56** plants in his garden this year.

Question Difficulty: Medium

Question ID 735a0a00

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 735a0a00

PSAT1 M 2.21

$$y = 0.25x^2 - 7.5x + 90.25$$

The equation gives the estimated stock price y , in dollars, for a certain company x days after a new product launched, where $0 \leq x \leq 20$. Which statement is the best interpretation of $(x, y) = (1, 83)$ in this context?

- A. The company's estimated stock price increased **\$83** every day after the new product launched.
- B. The company's estimated stock price increased **\$1** every **83** days after the new product launched.
- C. **1** day after the new product launched, the company's estimated stock price is **\$83**.
- D. **83** days after the new product launched, the company's estimated stock price is **\$1**.

ID: 735a0a00 Answer

Correct Answer: C

Rationale

Choice C is correct. In the given equation, x represents the number of days after a new product launched, where $0 \leq x \leq 20$, and y represents the estimated stock price, in dollars, for a certain company. Therefore, the best interpretation of $(x, y) = (1, 83)$ in this context is that **1** day after the new product launched, the company's estimated stock price is **\$83**.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 271ffad7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■ ■ ■

ID: 271ffad7

PSAT1 M 2.22

A quadratic function models a projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. The model estimates that the projectile was launched from an initial height of **7** meters above the ground and reached a maximum height of **51.1** meters above the ground **3** seconds after the launch. How many seconds after the launch does the model estimate that the projectile will return to a height of **7** meters?

- A. **3**
- B. **6**
- C. **7**
- D. **9**

ID: 271ffad7 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that a quadratic function models the projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. It follows that an equation representing the model can be written in the form $f(x) = a(x - h)^2 + k$, where $f(x)$ is the projectile's estimated height above the ground, in meters, x seconds after the launch, a is a constant, and k is the maximum height above the ground, in meters, the model estimates the projectile reached h seconds after the launch. It's given that the model estimates the projectile reached a maximum height of **51.1** meters above the ground **3** seconds after the launch. Therefore, $k = 51.1$ and $h = 3$. Substituting **51.1** for k and **3** for h in the equation $f(x) = a(x - h)^2 + k$ yields $f(x) = a(x - 3)^2 + 51.1$. It's also given that the model estimates that the projectile was launched from an initial height of **7** meters above the ground. Therefore, when $x = 0$, $f(x) = 7$. Substituting **0** for x and **7** for $f(x)$ in the equation $f(x) = a(x - 3)^2 + 51.1$ yields $7 = a(0 - 3)^2 + 51.1$, or $7 = 9a + 51.1$. Subtracting **51.1** from both sides of this equation yields $-44.1 = 9a$. Dividing both sides of this equation by **9** yields $-4.9 = a$. Substituting -4.9 for a in the equation $f(x) = a(x - 3)^2 + 51.1$ yields $f(x) = -4.9(x - 3)^2 + 51.1$. Therefore, the equation $f(x) = -4.9(x - 3)^2 + 51.1$ models the projectile's height, in meters, above the ground x seconds after it was launched. The number of seconds after the launch that the model estimates that the projectile will return to a height of **7** meters is the value of x when $f(x) = 7$. Substituting **7** for $f(x)$ in $f(x) = -4.9(x - 3)^2 + 51.1$ yields $7 = -4.9(x - 3)^2 + 51.1$. Subtracting **51.1** from both sides of this equation yields $-44.1 = -4.9(x - 3)^2$. Dividing both sides of this equation by -4.9 yields $9 = (x - 3)^2$. Taking the square root of both sides of this equation yields two equations: $3 = x - 3$ and $-3 = x - 3$. Adding **3** to both sides of the equation $3 = x - 3$ yields $6 = x$. Adding **3** to both sides of the equation $-3 = x - 3$ yields $0 = x$. Since **0** seconds after the launch represents the time at which the projectile was launched, **6** must be the number of seconds the model estimates that the projectile will return to a height of **7** meters.

Alternate approach: It's given that a quadratic function models the projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. It's also given that the model estimates that the projectile was launched from an initial height of **7** meters above the ground and reached a maximum height of **51.1** meters above the ground **3** seconds after the launch. Since the model is quadratic, and quadratic functions are symmetric, the model estimates that for any given height less than the maximum height, the time the projectile takes to travel from the given height to the maximum height is the same as the time the projectile takes to travel from the maximum height back to the given height. Thus, since the model estimates the projectile took **3** seconds to travel from **7** meters above the ground to its maximum height of **51.1** meters above the ground, the model also estimates the projectile will take **3** more seconds to travel from its maximum height of **51.1** meters above the ground back to **7** meters above the ground. Thus, the model estimates that the projectile will return to a height of **7** meters **3** seconds after it reaches its maximum height, which is **6** seconds after the launch.

Choice A is incorrect. The model estimates that **3** seconds after the launch, the projectile reached a height of **51.1** meters, not **7** meters.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard